Lesson for the course “Human-Computer Interaction”

The process of Interaction Design, Usability, and User Experience

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What is “user experience”?

User experience (UX)

- The User Experience is a complex **psychological response** to the interaction with computing systems
- This response is a consequence of
  - **Individual predispositions of the user**
    - (e.g., attitudes, motivations and needs)
  - **Characteristics of the interactive systems**
    - (e.g., purpose, functionality and usability)
  - **Contextual dependencies**
    - (e.g., task and environment)

User experience

- *"The User Experience encompasses **all aspects** of the end-user's interaction with the company, its services, and its products."* - Nielsen and Norman (2014)

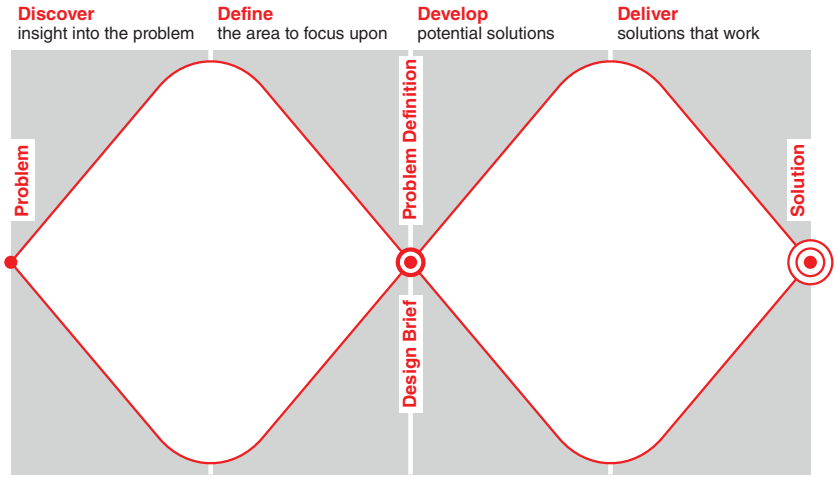
- Various aspects, including
  - **Pragmatic aspects**: how simple, practical and obvious is for the users to achieve their goals
  - **Hedonic aspects**: how evocative and stimulating the interaction is to the users

How can we design for quality user experiences?

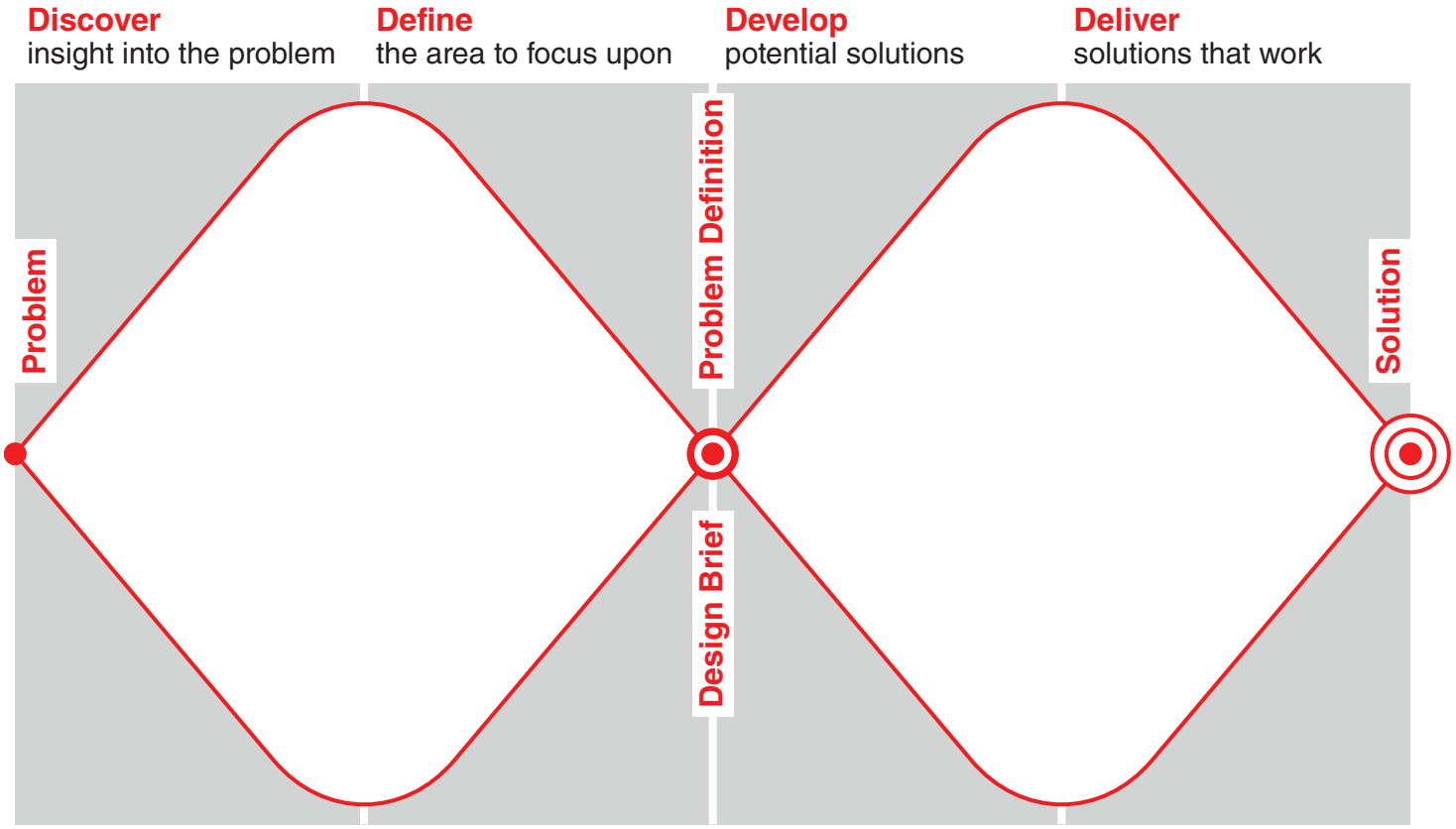
- There is no secret magical formula that can be readily applied by interaction designers…
- However, there are numerous
  - Conceptual frameworks
  - Tried and tested design methods
  - Guidelines
  - Relevant research findings
- These are described throughout this course
- **The key is understanding the users**

Design

- There are many fields of design, such as graphic design, architectural design, industrial design, and software design
- Although each discipline has its own approach to design, there are **commonalities**
- The Design Council of the United Kingdom captures these in **the double-diamond of design**
- This approach has 4 phases which are **iterated**:
  - **Discover**
  - **Define**
  - **Develop**
  - **Deliver**



The double-diamond of design



- **Discover**: Designers try to gather insights about the problem
- **Define**: Designers develop a clear summary that frames the design challenge
- **Develop**: Solutions or concepts are created, prototyped, tested, and iterated
- **Deliver**: The resulting project is finalized, produced, and launched

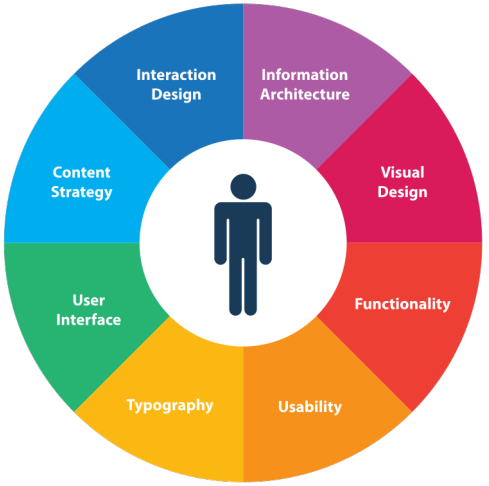
Interaction design (IxD)

- **IxD follows the 4 phases of the double-diamond**
- IxD it is underpinned by the philosophy of **user-centered design**
  - That is, involving users throughout development
  - IxD focuses on users since the very beginning of the design process
- Goal of IxD: to develop **usable** products as well as an **enjoyable user experience**
  - Usability
  - UX
  - …are not the same!

Interaction design

- It is a **process**:
  - Focused on discovering requirements, designing to fulfil requirements, producing prototypes and evaluating them
  - Focused on users and their goals
  - Involves trade-offs to balance conflicting requirements
  - Generating alternatives and choosing between them is key
- Is a **creative** activity (proposes new solutions)
- It is an iterative activity
  - Iteration is inevitable: **designers never get it right first time**

User-Centered Design approach



- The real users and their goals, not just technology, are the driving force behind product development

- The UCD approach is based on:
  - **Early focus on users and tasks:** understand the users by directly studying their cognitive, behavioral, anthropomorphic & attitudinal characteristics
  - **Empirical measurement:** users' reactions and performance to scenarios, simulations & prototypes are observed, recorded and analyzed
  - **Iterative design:** when problems are found in user testing, fix them and carry out more tests

Why is important to involve users in the design, development and evaluation ?

Importance of involving users

- Involving users in the design, development and evaluation process of a product is important because it is **the best way to ensure that the end product is usable and that it indeed will be used**
- They will tell you what are the right **requirements** for your product

4 basic activities of Interaction Design

1. **Discovering requirements** for the interactive product
2. **Designing alternatives** that meet those requirements
3. **Prototyping alternative designs** so that they can be communicated and assessed
4. **Evaluating product and its user experience** throughout the process

- These 4 phases are iterated

Four basic activities of Interaction Design

1. Discovering requirements

- This activity covers the left side of the double diamond of design (Problem Space)
- It is focused on discovering something new about the world and defining **what** will be developed
- This includes understanding the **target users** and the **support** an interactive product could usefully provide
- This understanding is discovered through **data gathering** and **analysis**
- This understanding forms the basis of the product's **requirements** and underpins subsequent design and development

Four basic activities of Interaction Design

2. Designing alternatives

- This is the core activity of designing and is part of the Develop phase of the double diamond (Solution Space): proposing ideas for meeting the requirements

- 2 sub-activities:
  - **Conceptual design**
    - Involves producing the conceptual model for the product
    - A conceptual model describes an abstraction outlining what people can do with a product and what concepts are needed to understand how to interact with it
  - **Concrete design**
    - Considers the detail of the product including the colors, sounds, and images to use, menu design, and icon design, etc.

- Alternatives are considered at every point

Four basic activities of Interaction Design

3. Prototyping alternative designs

- Prototyping is also part of the Develop phase of the double diamond (Solution Space)
- Interaction design involves designing
  - the **behavior** of interactive products
  - as well as their **look and feel**
- The **most effective way** for users to evaluate such designs is **to interact with them**, and this can be achieved through prototyping
  - E.g., effective for identifying problems in the early stages of design

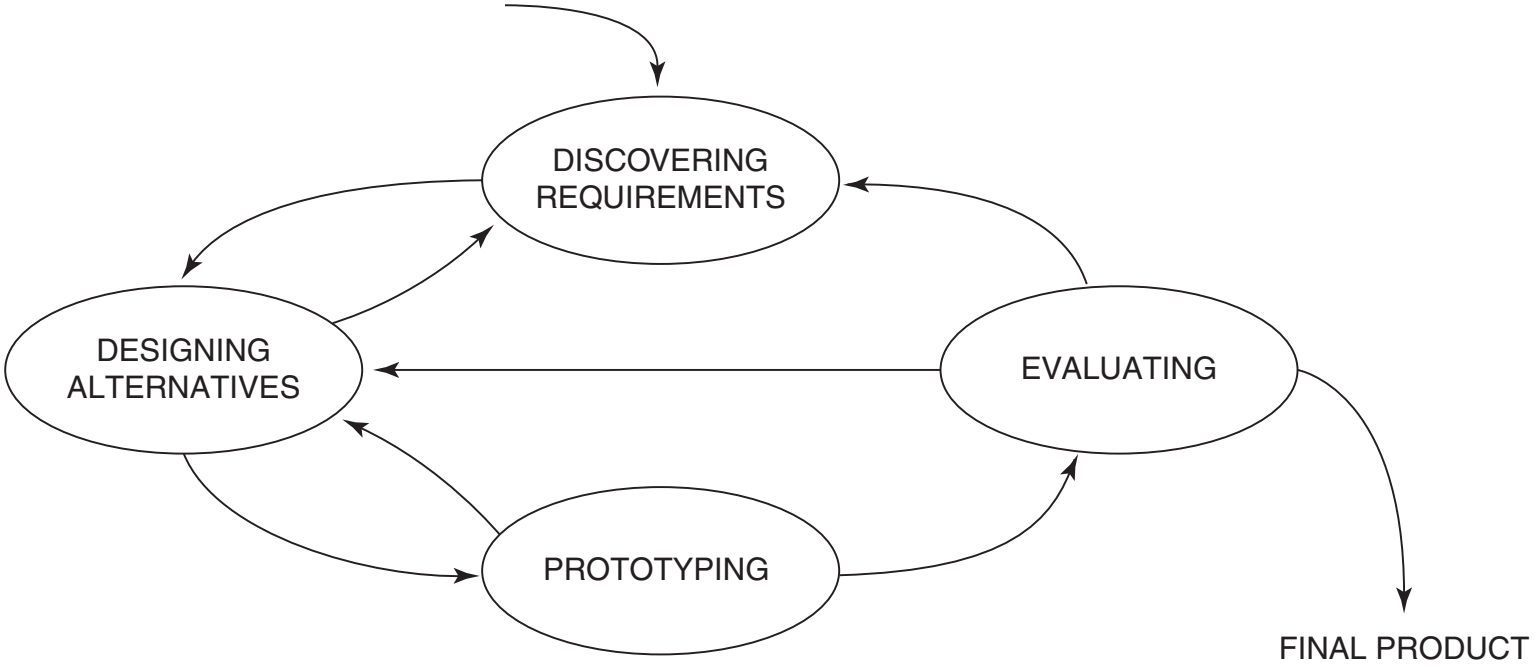
Four basic activities of Interaction Design

4. Evaluating

- Evaluating is also part of the Develop phase of the double diamond (Solution Space)
- It is the process of determining the success of a design, **measured** in terms of a variety of **usability and user-experience criteria**
- The evaluation phase allows you to choose among the alternative designs you have created

A simple interaction design lifecycle model

- The activities to discover requirements, design alternatives, build prototypes, and evaluate them are **intertwined**: alternatives are evaluated through the prototypes, and the results are fed back into further design or to identify alternative requirements



Understanding the problem space

- To design for a good experience, **it is important not to jump into decisions from the start**
  - Take a time to ask questions, find the answers, create user stories and explore the web for solutions to similar problems

- In the words of Don Norman, *"a brilliant solution to the wrong problem can be worse than no solution at all"*

- **Good designers never start by trying to solve the problem given to them: they start by trying to understand what the real issues are**



Understanding the problem space

- Designers need to understand the people **who** will use their systems and products
- They need to understand the **activities** that people want to undertake and the **contexts** in which those activities take place
- Designers also need to know about the features of interactive **technologies** and how to approach designing interactive systems

Understanding the problem space

- Explore:
  - What is currently the user experience
  - Why is a change needed
  - How will this change improve the situation

- Articulating the problem space is typically done as a **team effort**
  - This allows
    - **To explore different perspectives**
    - **To avoid incorrect assumptions and unsupported claims**



Understanding the problem space

- E.g., AR displays and holographic navigation systems that are available in some cars nowadays
  - They are the result of decades of research into human factors
  - Understanding the problem space has been critical in arriving at workable solutions that are safe and trusted



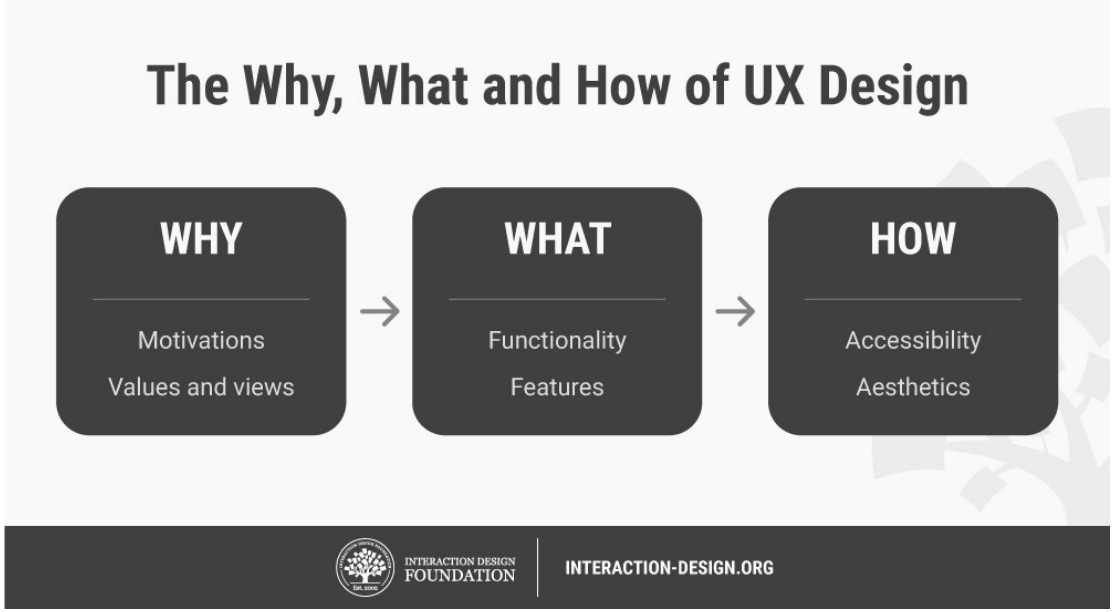


UX Designers consider the Why, What and How of product use

- The **Why** involves the users' motivations for adopting a product, whether they relate to a task they wish to perform with it or to values and views which users associate with the ownership and use of the product
- The **What** addresses the things people can do with a product – its functionality
- The **How** relates to the design of functionality in an accessible and aesthetically pleasant way

UX Designers consider the Why, What and How of product use

- UX designers start with the Why before determining the What and then, finally, the How in order to create products that users can form meaningful experiences with



- Now we address the What: the concept of usability

What is “usability”?

Usability

- A central concern of interaction design is to develop interactive products that are **usable**
- Usability is an important **quality indicator** for interactive systems
- Usability refers to the **degree** to which products are easy to use, easy to learn, effective, efficient, safe to errors and satisfying to users
- Usability is mainly about the **functional part** of the product: it addresses whether or not you are able to achieve a task or goals with a product
- The ISO 9241-11 standard on usability describes it as: *"The extent to which a product can be used by specified users to achieve specified goals, with effectiveness, efficiency and satisfaction in a specified context of use."*

The importance of usability

- Usability matters because if users cannot achieve their goals efficiently, effectively and in a satisfactory manner, they are likely to seek an alternative solution to reach their goals
  - and for websites and apps, alternative solutions are abundant
- Quite simply: if your product is not usable, its UX will be bad, and users will **leave** you for your **competitors**



Usability dimensions/goals

- Usability can be defined in terms of the following dimensions/goals:
  - Effective to use (**effectiveness**)
  - Efficient to use (**efficiency**)
  - Safe to use (**safety**)
  - Having good utility (**utility**)
  - Easy to learn (**learnability**)
  - Easy to remember how to use (**memorability**)

Usability dimensions are typically put into practice as questions

- The purpose of the questions is to provide the interaction designer with concrete means of **assessing** various aspects of an interactive product
- Through answering the questions, designers can be alerted **very early on in the design process** to potential design problems and conflicts that they might not have considered

Usability dimensions are typically put into practice as questions

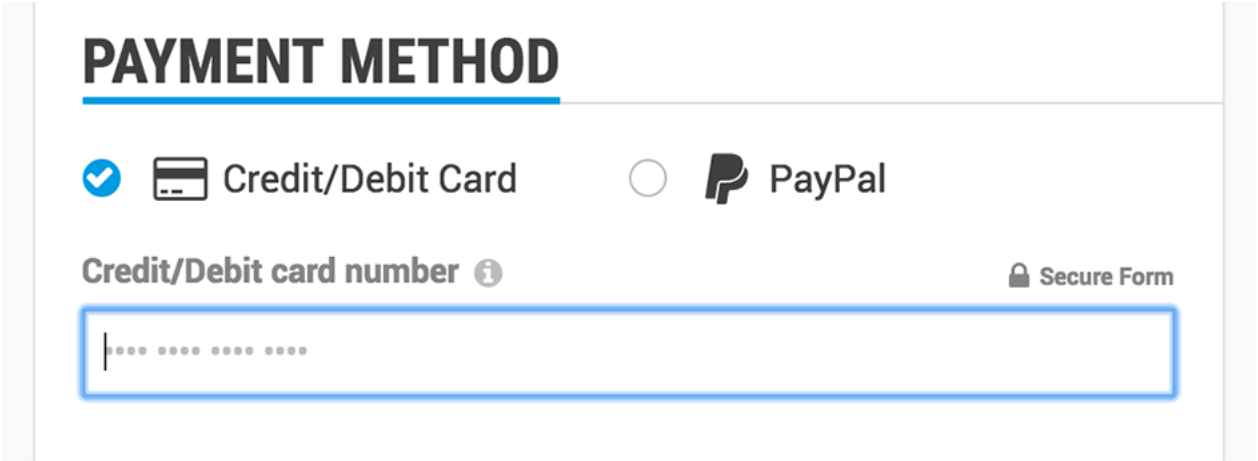
- However, simply asking "Is the system easy to learn?" is not going to be very helpful…
- Asking about the usability of a product in a **more detailed way** will elicit far more information.
- E.g.,
  - How long will it take a user to figure out how to use the most basic functions for a new smartwatch?
  - How much can users capitalize on their prior experience?
  - How long would it take the user to learn the whole set of functions?

Effectiveness

- It refers to **how good a product is at doing what it is supposed to do**
- Effectiveness is about whether users can complete their goals with a high degree of accuracy
  - How completely and accurately the task is completed or goals reached
- Much of the effectiveness of a product comes from the **support** provided to users when they work with the product
  - The key is to be as *informative* as possible in a *meaningful way* to the user

Effectiveness

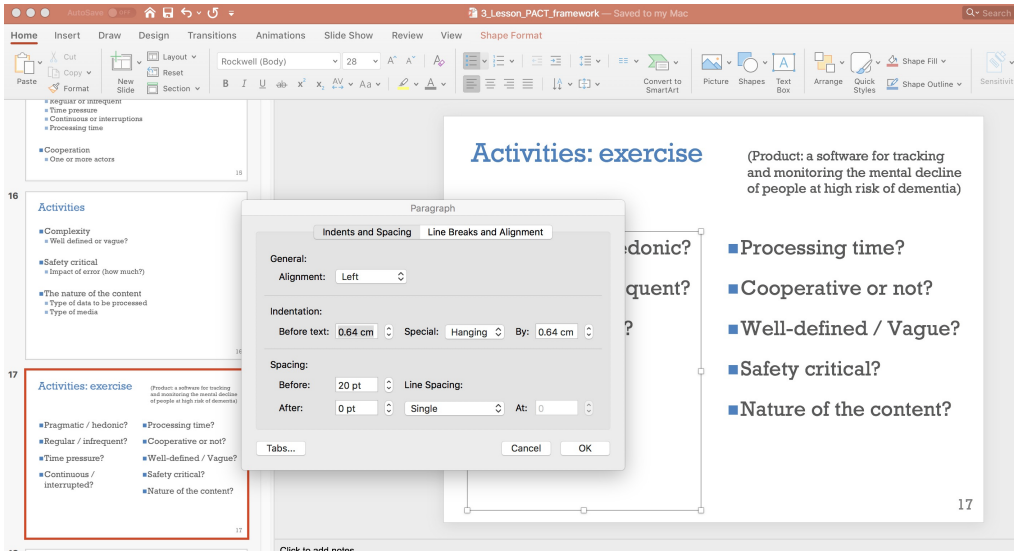
- Example

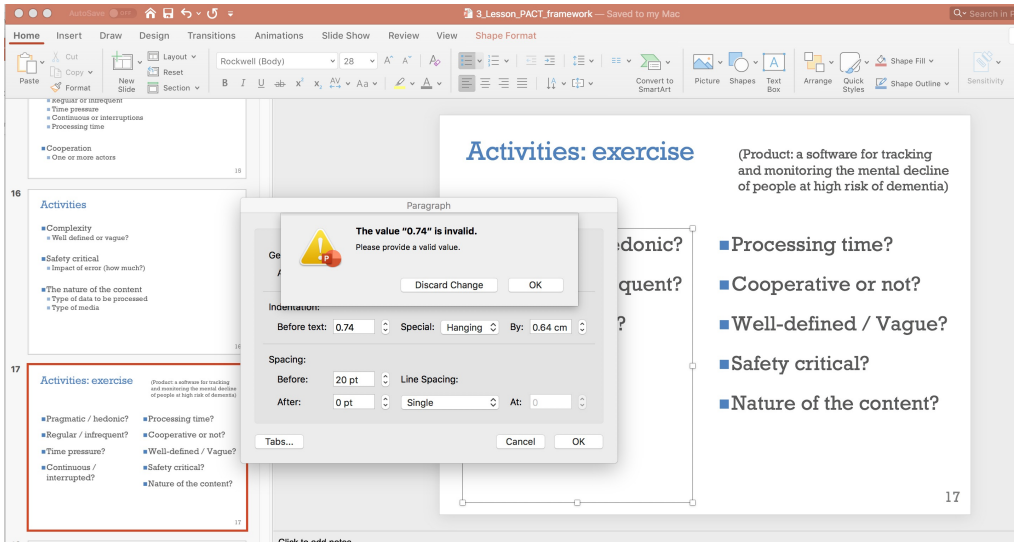


A credit card field that **only** accepts a valid credit card number entry can reduce data entry errors and help users perform their tasks correctly

- **Question:** Is the product capable of allowing people to learn, carry out their work efficiently, access the information that they need, or buy the goods that they want?

Effectiveness





- The user has forgotten to place "cm" next to the number
- The error simply state "invalid" and there is no indication about what needs to be fixed
- A more usable interface should let the user choose the measure unity from a dropdown menu, and have a default value (i.e., the last used one, the one that the user wants to modify)

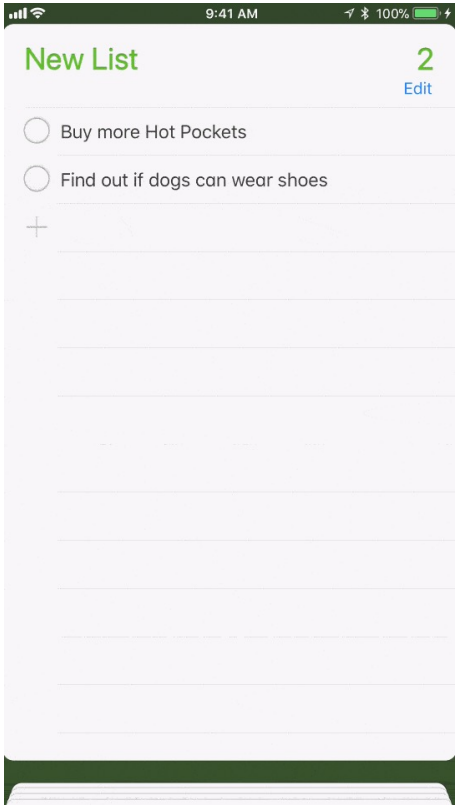
Efficiency

- It refers to **the way a product supports users in carrying out their tasks**
- Efficiency is all about **speed**. *How fast can the user get the job done?*
  - You'll want to examine the number of steps (or clicks/keystrokes) to achieving the objective; can they be reduced? This will help develop efficient processes
- **Question:** Once users have learned how to use a product to carry out their tasks, can they sustain a high level of productivity? How quickly the task can be completed?

Efficiency: compare 2 UIs

- Creating reminders: text-based app vs SIRI

1. Open the app
2. Tap a new line
3. Type out the reminder
4. Tap the calendar button
5. Select the date and time
6. Tap "Done"



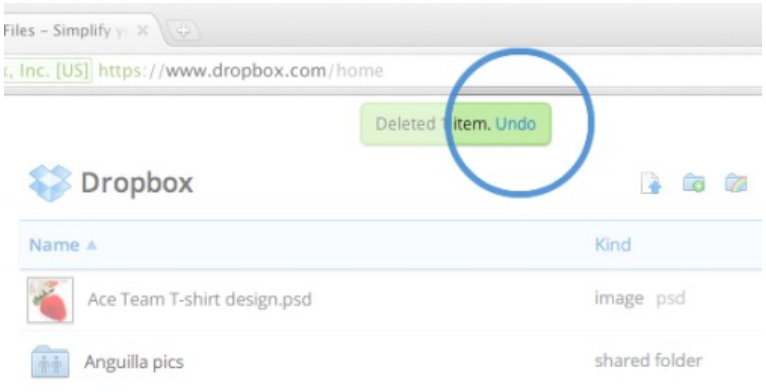


"Hey Siri, remind me to email my mom tomorrow morning 10 AM"

Safety

- It refers to **helping any kind of user in any kind of situation to avoid the dangers of carrying out unwanted actions accidentally**

- Making interactive products safer in this sense involves
  1. **Preventing** the user from making serious **errors** by reducing the risk of wrong keys/buttons being mistakenly activated (e.g., don't place the quit or delete-file command right next to the save command on a menu)
  2. **Providing** users with various **means of recovery** should they make errors, such as an undo/redo functions



Safety

- Safe interactive systems should generate confidence and allow the user to explore the interface to try new operations
- Example of safety mechanism is confirming dialog boxes that give users another chance to consider their intentions



- **Question:** What is the range of errors that are possible using the product, and what measures are there to permit users to recover easily from them?

Utility

- It refers to **the extent to which the product provides the right kind of functionality so that users can do what they need or want to do**
- Utility is about providing functions that users *need* in the first place
  - Don't provide features in your app/website/program that users don't need
- **Question:** Does the product provide an appropriate set of functions that will enable users to carry out all of their tasks in the way they want to do them?

Learnability

- It refers to **how easy a system is to learn to use**
- **Question:** Is it possible for the user to work out how to use the product by exploring the interface and trying certain actions? How hard will it be to learn the whole set of functions in this way?

Memorability

- It refers to **how easy a product is to remember how to use, once learned**
  - This is especially important for interactive products that are used infrequently
  - If users haven't used an operation for a few months or longer, they should be able to remember or at least rapidly be reminded how to use it
  - Users shouldn't have to keep relearning how to carry out tasks
  - Unfortunately, this tends to happen when the operations required to be learned are obscure, illogical, or poorly sequenced

Memorability

- Users need to be helped to remember how to do tasks
- **Question:** What types of interface support have been provided to help users remember how to carry out tasks, especially for products and operations they use infrequently?

- In a few weeks we will have a lesson about memory

Usability criteria

- Usability dimensions can be turned into **usability criteria**
- These are specific objectives that enable the usability of a product to be assessed in terms of how it can improve (or not improve) a user's performance

Usability criteria

- Examples of commonly used usability criteria are
  - Time to complete a task (efficiency)
  - Time to learn a task (learnability)
  - Number of errors made when carrying out a given task over time (memorability/safety)

- However, **they do not address the overall quality of the user experience**, which is where **user experience dimensions** come into play

User experience dimensions

- **Desirable aspects**
  - Satisfying
  - Fun
  - Enjoyable
  - Entertaining
  - Helpful
  - Motivating
  - Aesthetically pleasing
  - Enhancing sociability
  - Rewarding
  - Support creativity
  - Emotionally fulfilling
- **Undesirable aspects**
  - Boring
  - Unpleasant
  - Frustrating
  - Making one feel guilty
  - Making one feel stupid
  - Annoying
  - Patronizing
  - Childish

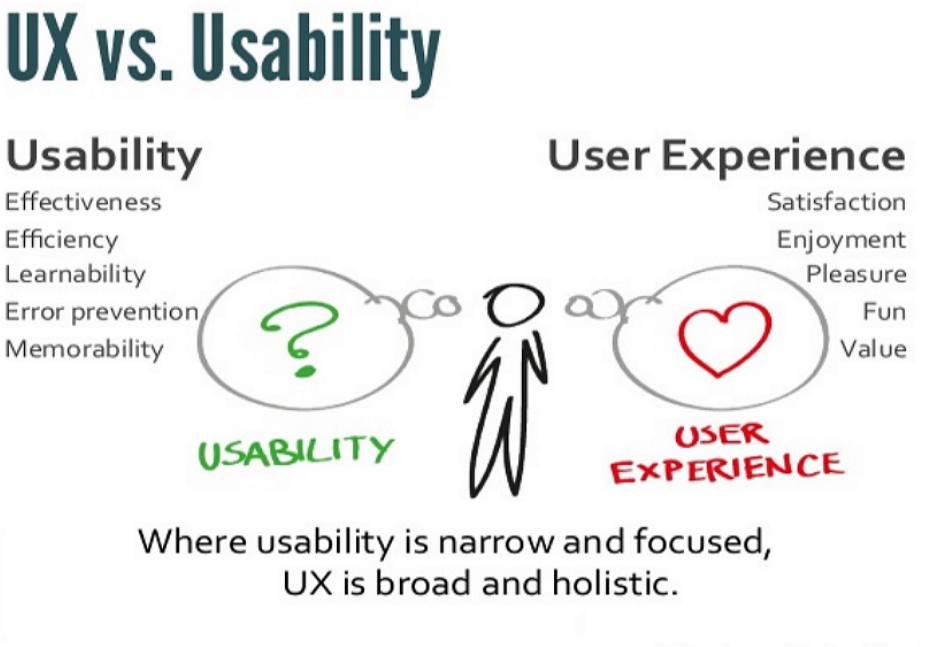
User experience dimensions

- Many of these desirable and undesirable aspects are **subjective qualities** and are concerned with how a system **feels** to a user

- They differ from the **more objective usability dimensions** in that they are concerned with how users experience an interactive product **from their perspective**, rather than assessing how useful or productive a system is from its own perspective

Usability and user experience are not the same thing

- Usability is a quality attribute of a product. It is **mainly about the functional part of the product**
  - The usability of a product is a crucial part that shapes its UX
- **User experience** refers to a person's subjective feeling and attitudes about using the product



Usability and user experience are not the same thing

- Usability involves effectiveness, efficiency, and satisfaction where users can complete certain **tasks** as easy as possible
- On the other hand, user experience covers a **wider** aspect of the user's interaction with the product
- Usability relates to the **ease with which users can achieve their goals** while interacting with a website while user experience is concerned with **the way users perceive their interaction** with that product

The term UX according to its inventor



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Don Norman: The term "UX"

176.574 visualizzazioni

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We asked Don what he feels about the way people are using terms like “UX” and “user experience” these days.

https://youtu.be/9BdtGjoIN4E

Usability testing

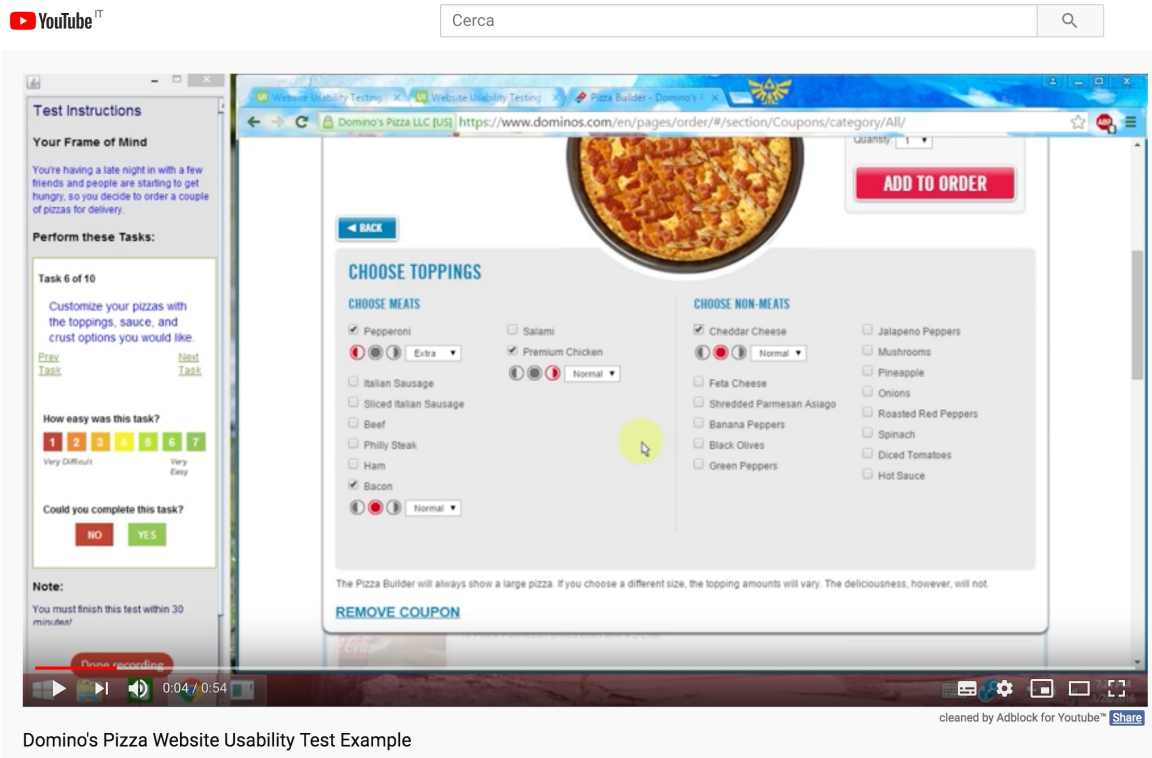
- Usability testing is the practice of testing how easy a design is to use on a group of **representative** users

- It usually involves **observing users** as they attempt to complete tasks and can be done for different types of designs, from user interfaces to physical products

- It is often **conducted repeatedly**, from early development until a product's release

Usability testing example

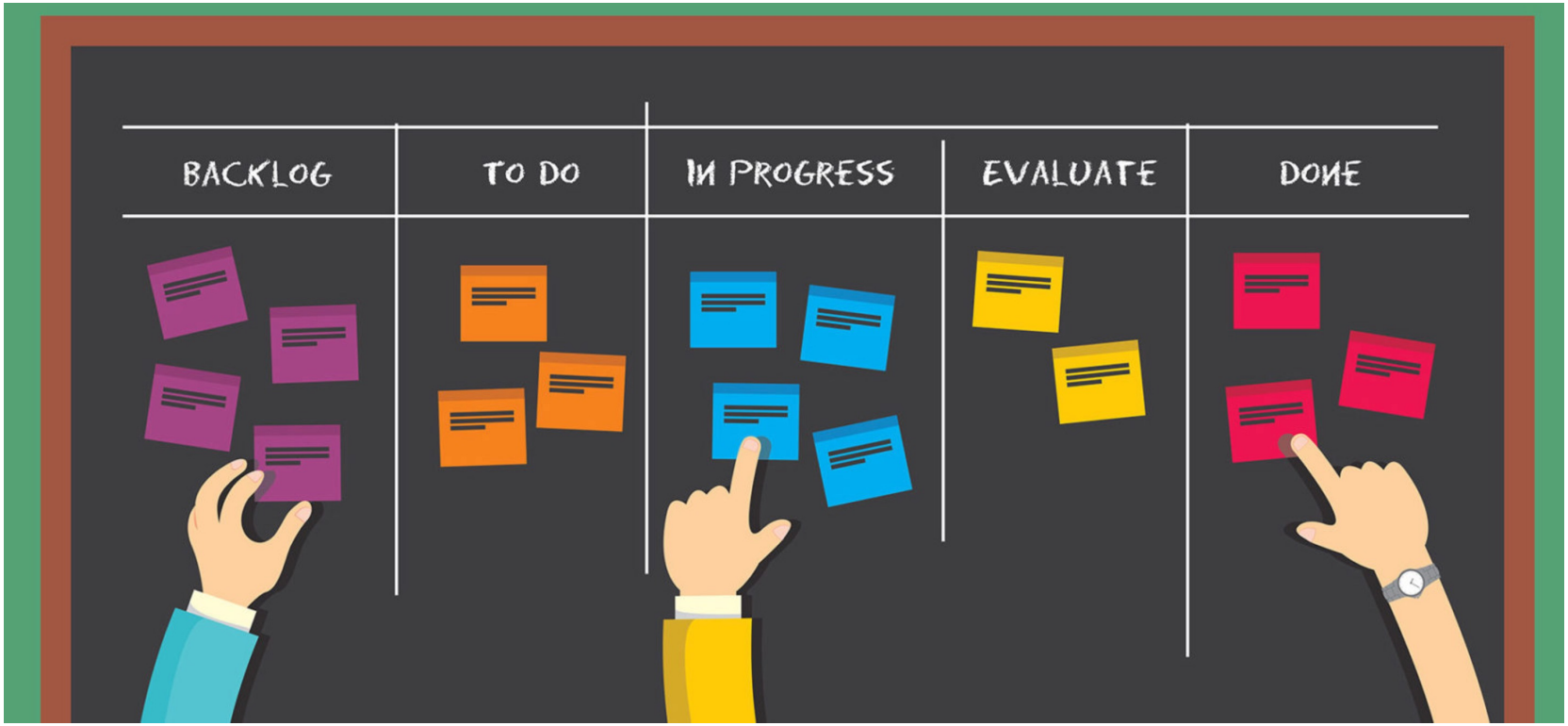
- https://youtu.be/2IRRm3_Ywnw



Domino's Pizza Website Usability Test Example

- Usability testing may reveal design problems that frustrate and confuse users, like the 2 buttons this user couldn't choose between while ordering her pizza, in this video

Project management



Agile development and Scrum

- Agile software development consists of:
  - Short timeboxes of iterative development (sprint, iteration, cycle)
  - Early and repeated customer/user feedback
  - Re-prioritisation of work based on customer or user, so that emergent requirements can be handled
- Agile development arose against heavily planned and regulated approaches to developing software
- Many different types of Agile method implementations exist, e.g., **Scrum**

Scrum

- Is an agile way to **manage** a project
- The concept is to break large projects into smaller stages, reviewing and adapting along the way
- Scrum relies on a self-organizing, cross-functional team
  - **Self-organizing**: there is no overall team leader who decides which person will do which task or how a problem will be solved. Those are issues that are decided by the team as a whole
  - **Cross functional**: everyone is needed to take a feature from idea to implementation

Scrum

- Is based on **continuous learning and adjustment to fluctuating factors**
- It acknowledges that the team doesn't know everything at the start of a project and will evolve through experience
- It is structured to help teams naturally **adapt to changing conditions and user requirements**, with re-prioritization built into the process and short release cycles

Scrum: components of the framework

- **Product Backlog**
  - Is an ordered list of everything that is known to be needed in a product (it is, essentially, the team's **"To Do" list**)
  - It is constantly evolving and is never complete
- **Sprint**
  - Is a short, time boxed period (no more than a month long, most commonly two weeks) when a scrum team works to complete a set amount of work

Scrum

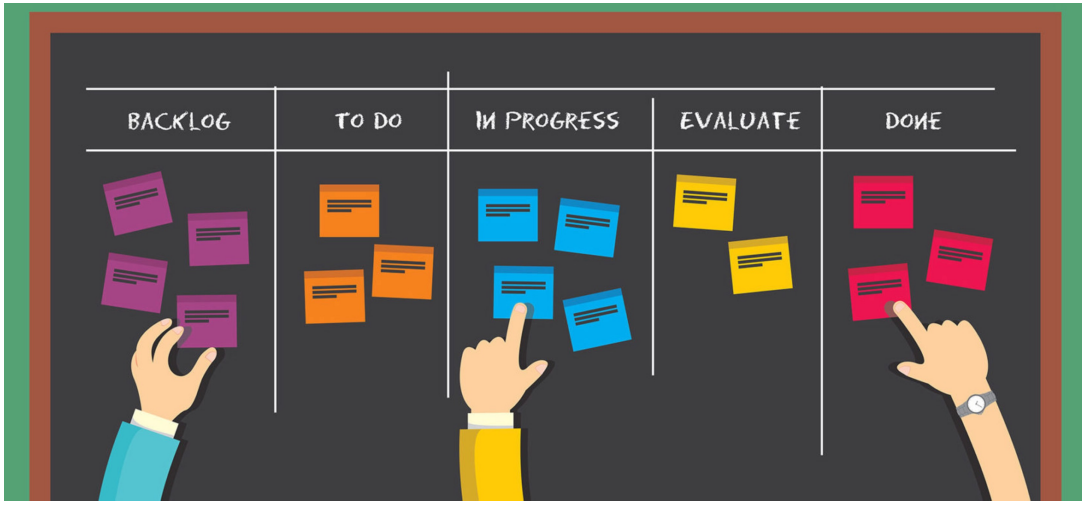
- There are 4 events that happen with each Sprint:

1. **Sprint Planning**: the team decides what to work on for the current period
   - A Sprint Backlog is created: it is the list of items selected (from the Product Backlog) for implementation in the current sprint cycle
2. **Daily Scrum**: the team meets for 15 minutes (or less) every day of the Sprint to inspect progress toward the Sprint Goal
3. **Sprint Review**: the team collaborates about what was done and adapts the Backlog as needed
4. **Sprint Retrospective**: the team discusses what went right, what went wrong, and how to improve

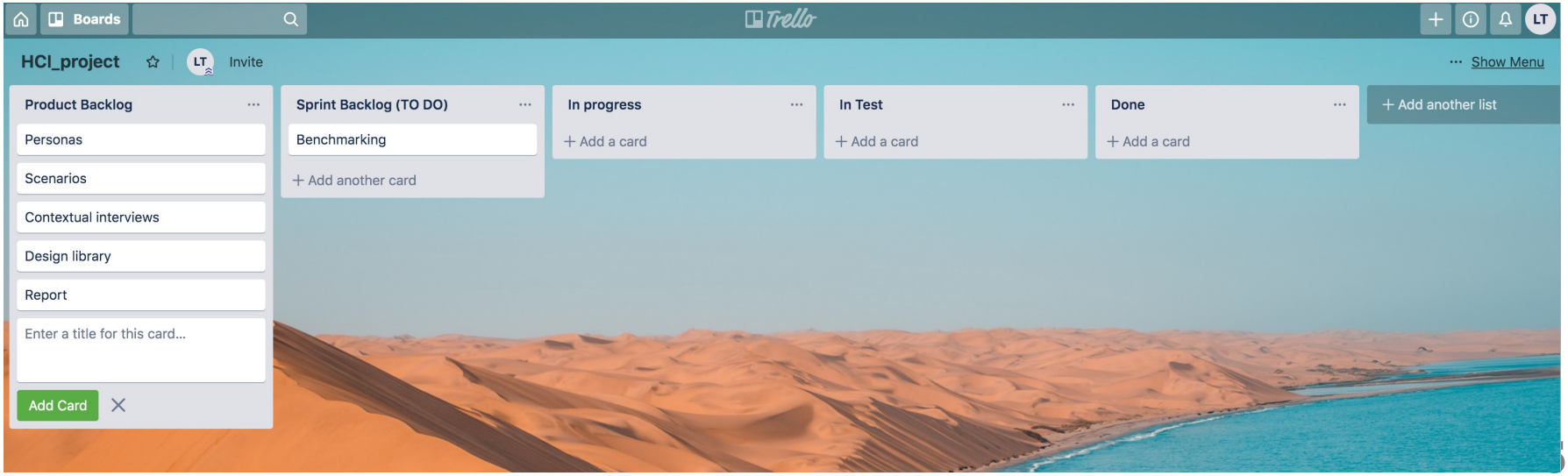
Scrum: roles

- Within agile development, Scrum teams are supported by two specific roles
- **Scrum Master**
  - can be thought of as a **coach** for the team, helping team members use the Scrum process to perform at the highest level
  - also protects the team from both internal and external distractions
- **Product Owner**
  - maintains the Product Backlog, ensures that everyone knows the priorities, and is expected to do the best possible job of satisfying all stakeholders

Software that support the Scrum framework



- Trello: https://trello.com/



Homework

- Select a website, an app, and a program and assess their usability
  - Write down some notes
- Also, reflect on your experience in interacting with those products
  - How do you feel?
  - Write down some notes
- Create an account on Trello and start using it to manage your project

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